Design Phase Report: Polyline Editor

In this phase, the system transitions from understanding user needs (analysis) to defining how the application will function in practice using HCI principles.

1. **Structural Choice & Notation:** The system is designed using State Transition Networks (STN) to ensure a deterministic flow. Each state such as IDLE, DRAWING, MOVING, and DELETING is clearly defined. Transitions like pressing 'b', 'm', or mouse clicks ensure the system never gets stuck. Task flows are also structured, such as adding vertices step-by-step during drawing mode.

2. **Visibility & Feedback (Heuristics):** The design ensures strong user feedback. Selected vertices are highlighted using visual indicators such as a red glow or circle to avoid precision issues. A persistent status bar displays the current mode to reduce confusion and improve usability.

3. **Controls & Affordances:** User interaction is simplified through intuitive controls. Keyboard inputs like 'd' for delete and 'r' for refresh are mapped logically. Mouse interaction is optimized using a 10-pixel threshold based on Fitts's Law to balance accuracy and speed.

4. **Constraints & Safety Nets:** The system includes safeguards to prevent user errors. Undo functionality (Ctrl+Z) allows recovery from mistakes. Deleting a vertex automatically reconnects adjacent points, ensuring data integrity and maintaining the polyline structure.

Conclusion: The Design Phase successfully transforms requirements into a structured, user-friendly system. All major HCI principles such as consistency, feedback, error prevention, and recoverability have been implemented to ensure a robust and efficient Polyline Editor.

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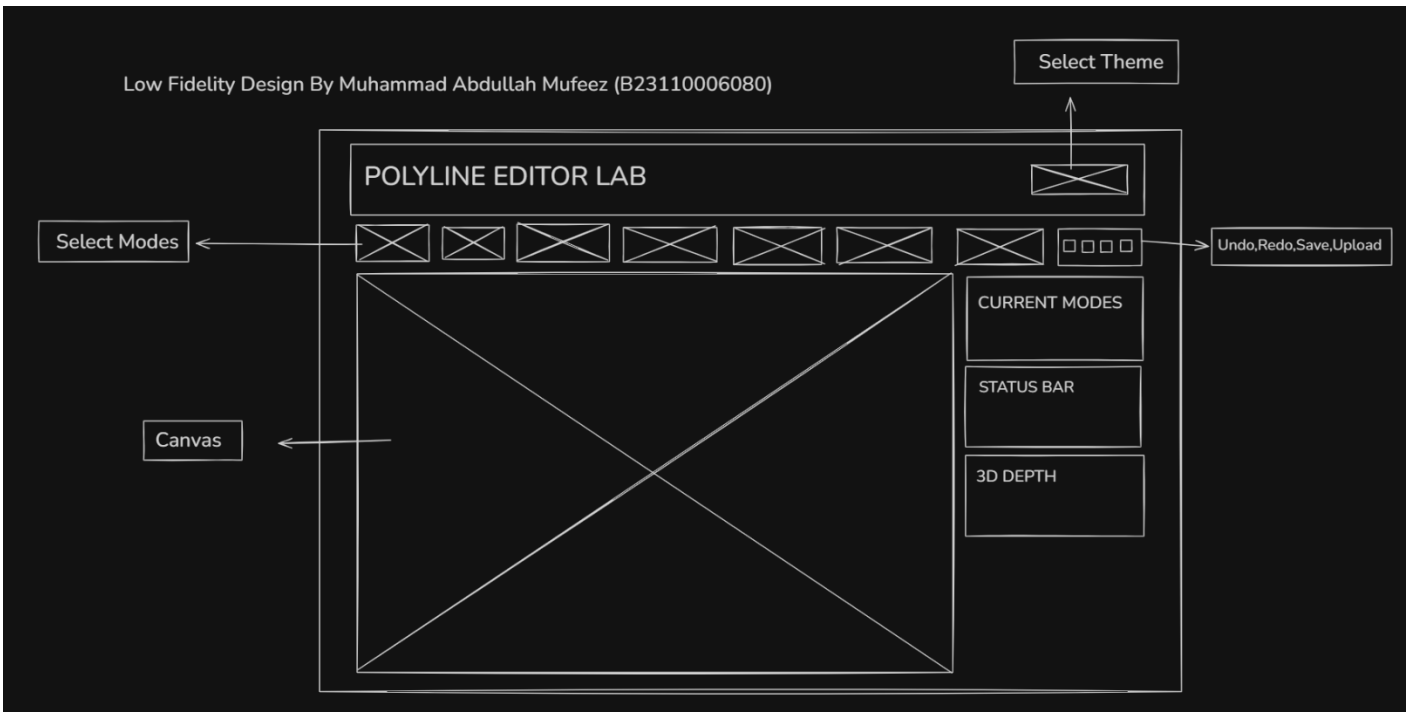
Low Fidelity Design



Select Modes Button:

- BEGIN – B
- DELETE – D
- MOVE – M
- HAND – H
- REFRESH – R
- QUIT – Q
- INSERT – I

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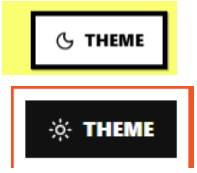
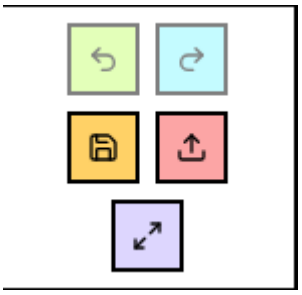
Select Theme:

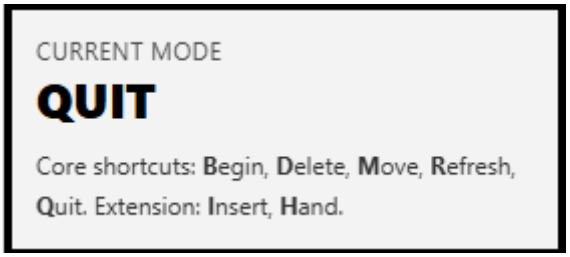
- Dark
- Light



Save, Undo, Redo, Upload, Zoom Button:

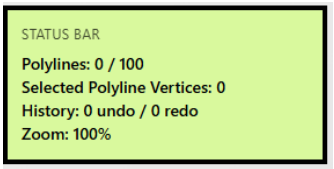


CURRENT MODES: Displays Current Mode



STATUS BAR:

Tells the number of Polylines with limit, selected polylines vertices, history of undo and redos, and Zoom ratio.



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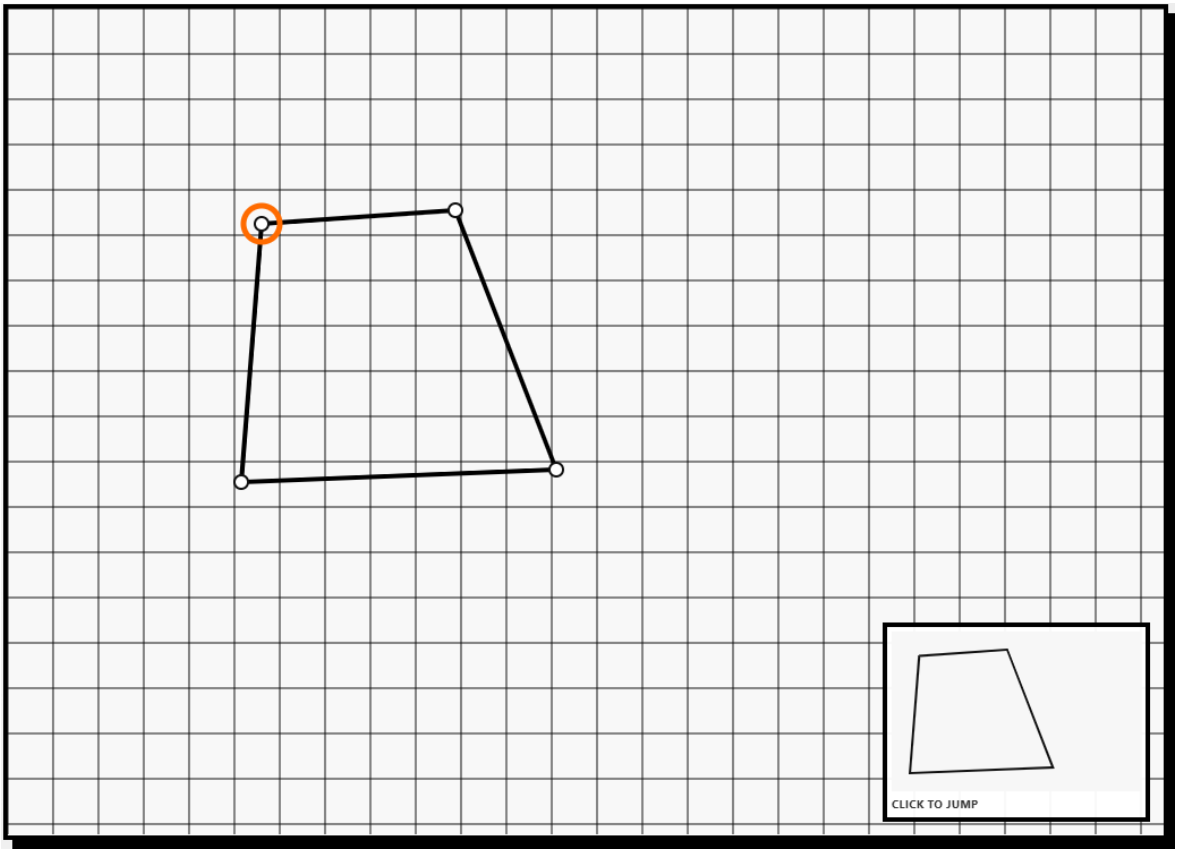
3D Depth:

This is the name of the setting. It indicates you are adjusting how "deep" or "far back" a point is in 3D space.

Canvas:

This is where the user will draw lines.



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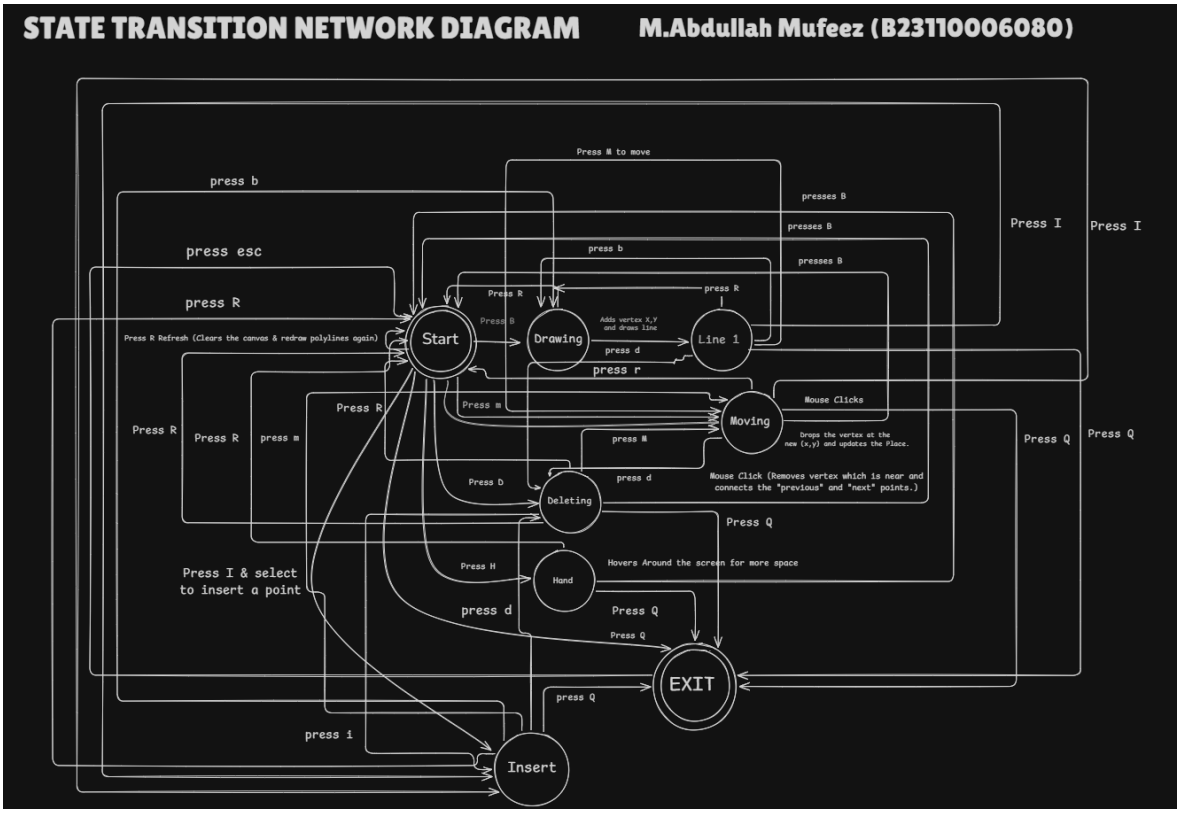
State Transition Network (STN)



Begin --> Begin: click canvas (add vertex)
Begin --> Delete: key d
Begin --> Move: key m
Begin --> Refresh: key r
Begin --> Quit: key q
Begin --> Insert: key i

Delete --> Begin: key b
Delete --> Move: key m
Delete --> Insert: key i
Delete --> Refresh: key r
Delete --> Quit: key q

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Move --> Begin: key b
Move --> Delete: key d
Move --> Insert: key i
Move --> Refresh: key r
Move --> Quit: key q

Insert --> Begin: key b
Insert --> Delete: key d
Insert --> Move: key m
Insert --> Refresh: key r
Insert --> Quit: key q

Quit --> Begin: key b or Escape

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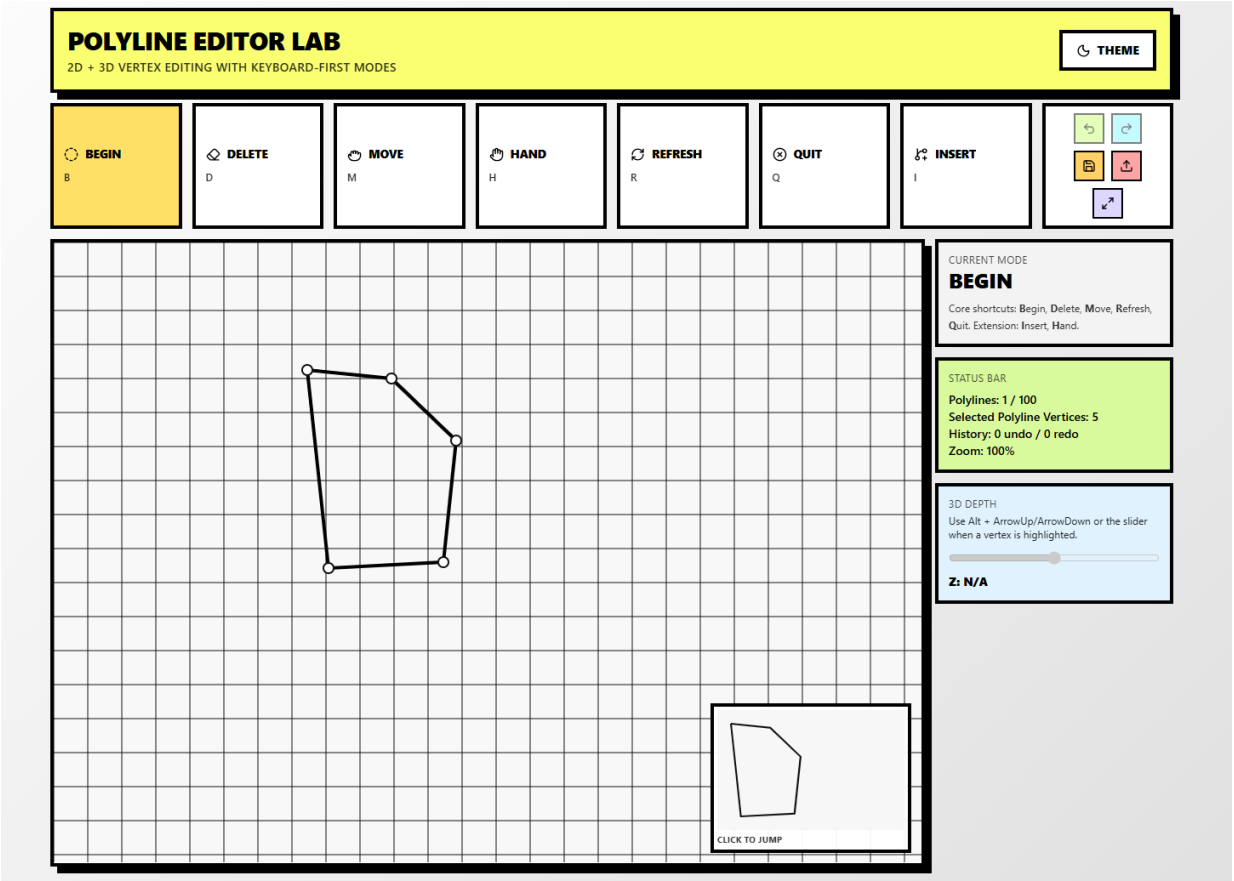
The Final Design



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